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Augmenting Human Cognition with Computers

Papert Lab

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How can we develop transformative tools for thought?

Andy Matuschak and Michael Nielsen

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- Might it be that one day most human beings will have a regular *memory practice*, as part of their everyday lives? Can we make it so memory becomes a choice; is it possible to in some sense solve the problem of memory?

How to invent Hindu-Arabic numerals?

Let's briefly get away from memory systems. Imagine you're a designer living in ancient Rome, working for MDC (Mathematical Designs Corporation). A client comes in one day, expressing a desire to improve on Roman numerals. Of course, that's not literally how they describe their problem to you – more likely it's a tax collector wanting to tabulate taxes more efficiently, and having some vague notion that MDC may be able to help. But to you, an experienced designer, it seems that an improved system of numerals may be what they need.

How should you respond to this request? From our modern vantage point we know a vastly better system of numerals is possible, the Hindu-Arabic numerals. Hindu-Arabic numerals were, in fact, a great leap in the history of tools for thought. Could you, as a designer, have made that leap? What creative steps would be needed to invent Hindu-Arabic numerals, starting from the Roman numerals? Is there a creative practice in which such steps would be likely to occur?

To be clear: this is a somewhat fanciful thought experiment. Many of the ideas needed to get to Hindu-Arabic numerals were, in fact, known earlier to the Babylonians, to the Greeks, and in other cultures. They were also inchoate in the abacus and simi-

lar devices. And so we're not asking a literal historical question. Rather, it's a question meant to stimulate thought: what *design process* could take you from Roman numerals to Hindu-Arabic numerals?

We can't know the answer to this question for sure. But it's worth pointing out that the Hindu-Arabic numerals aren't just an extraordinary piece of design. They're also an extraordinary mathematical insight. They involve many non-obvious ideas, if all you know is Roman numerals. Perhaps most remarkably, the meaning of a numeral actually *changes*, depending on its position within a number. Also remarkable, consider that when we add the numbers 72 and 83 we at some point will likely use $2+3=5$; similarly, when we add 27 and 38 we will also use $2+3=5$, despite the fact that the meaning of 2 and 3 in the second sum is completely different than in the first sum. In modern user interface terms, the numerals have the same affordances, despite their meaning being very different in the two cases. We take this for granted, but this similarity in behavior is a consequence of deep facts about the number system: commutativity, associativity, and distributivity. All these properties (and many more) point to the design and mathematical insights being inextricably entangled: the mathematical insights are, in some sense, design insights, and vice versa.

Indeed, it seems fair to say that any person who could invent Hindu-Arabic numerals, starting from the Roman numerals, would be both one of the great mathematical geniuses who ever lived, and one of the great design geniuses who ever lived. They'd have to be extraordinarily capable in both domains, capable of an insight-through-making loop which used the evolving system of numerals to improve not just their own mathematical ideas, but to have original, world-class insights into mathematics; and also to use those mathematical insights to improve their evolving system of numerals.

This is rather sobering if we compare to conventional modern design practice. In a typical practice, you'd interview domain experts (in this case, mathematicians), and read any relevant literature. You'd talk to users of existing systems, and analyze serious behavior, both individually and at scale. In short, you'd do what people in the design community refer to as immersing themselves in the target field.

This is a powerful practice. At its best it causes systems to come into existence which would otherwise be inconceivable. If applied to Roman numerals (in hypothetical ancient Rome, not today) this practice would likely improve them a great deal. But it would not provide anywhere near the level of mathematical insight needed to arrive at Hindu-Arabic numerals.

Our story about Hindu-Arabic numerals and mathematics is fanciful. But it expresses a general truth: *the most powerful tools for thought express deep insights into the underlying subject matter*. In the case of memory systems, this means they're not just “applied cognitive science”, a collage of existing ideas from cognitive science pasted together using modern design practice. Rather, they will express deep original insights into memory, insights no-one else in the world has ever had. A truly great memory system will be cognitive science of the highest order.

From this discussion, we take away a warning and an aspiration.

The warning is this: conventional tech industry product practice will not produce deep enough subject matter insights to create transformative tools for thought. Indeed, that's part of the reason there's been so little progress from the tech industry on tools for thought. This sounds like a knock on conventional product practice, but it's not. That practice has been astoundingly successful at its purpose: creating great businesses. But it's also what Alan Kay has dubbed a pop culture, not a research culture. To build transformative tools for thought we need to go beyond that pop culture.

The aspiration is for any team serious about making transformative tools for thought. It's to create a culture that combines the best parts of modern product practice with the best parts of the (very different) modern research culture. You need the insight-through-making loop to operate, whereby deep, original insights about the subject feed back to change and improve the system, and changes to the system result in deep, original insights about the subject.

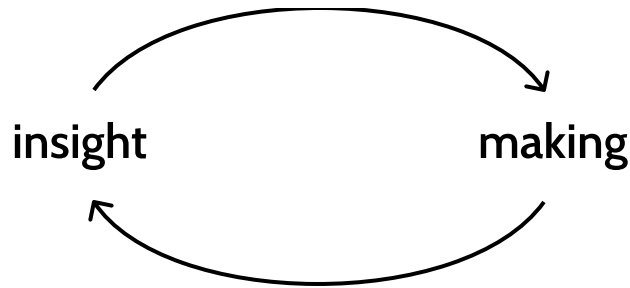


Diagram: 'insight' and 'making', pointing to each other in a loop

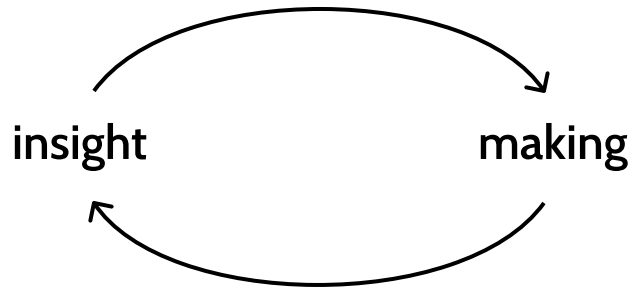


Diagram: 'insight' and 'making', pointing to each other in a loop

Note that we are not making the common argument that making new tools can lead to new subject matter insights for the toolmaker, and vice versa. This is correct, but is much weaker than what we are saying. Rather: making new tools can lead to new subject matter insights *for humanity as a whole* (i.e., significant original research insights), and vice versa, and this would ideally be a rapidly-turning loop to develop the most transformative tools.

Doing this is a cultural struggle. It seems to be extraordinarily rare to find the insight-through-making loop working at full throttle. People with expertise on one side of the loop often have trouble perceiving (much less understanding and participating in) the nature of the work that goes on on the other side of the loop. You have researchers, brilliant in their domain, who think of making as something essentially trivial, “just a matter of implementation”. And you have makers who don't understand research at all, who see it as merely a rather slow and dysfunctional (and unprofitable) making process. This is certainly true in Silicon Valley, where it's common to meet accomplished technical makers who, after reading a few stories from

Richard Hamming and Richard Feynman, think they understand research well enough that they can “create the new Bell Labs”. Usually they're victims of Dunning-Krugeritis, so ignorant they're not even aware of their ignorance.

Why isn't there more work on tools for thought today?

If tools for thought are so great, why isn't more work being done on them? Why aren't they a major industry?

As noted in the introduction, there's certainly a lot of lip service paid. It is, for instance, common to hear technologists allude to Steve Jobs's metaphor of computers as “bicycles for the mind”. But in practice it's rarely more than lip service. Many pioneers of computing have been deeply disappointed in the limited use of computers as tools to improve human cognition. Douglas Engelbart disparaged the “dangerous, disappointing, narrow, path that we seem to be stuck with following”. When asked in 2006 how much of his vision had been achieved, Engelbart replied facetiously “about 2.8 percent”. Alan Kay gives talks asserting “The real computer revolution hasn't happened yet” and in an interview has described the modern web as “reinventing the flat tire... at least give us what Engelbart did, for Christ's sake.”

Our experience is that many of today's technology leaders genuinely venerate Engelbart, Kay, and their colleagues. Many even feel that computers have huge potential as tools for improving human thinking. But they don't see how to build good businesses around developing new tools for thought. And without such business opportunities, work languishes.

What makes it difficult to build companies that develop tools for thought? To answer this, consider Adobe, one of the few large companies serious about developing new tools for thought. It's poured money into developing new mediums for designers and artists – programs such as *Illustrator*, *Photoshop*, and so on. These mediums are remarkable tools for thought.

Unfortunately for Adobe, such mediums are extremely expensive to develop, and it's difficult to prevent other companies from cheaply copying the ideas or developing near-equivalents. Consider, for example, the way the program *Sketch* has eaten into Adobe's market share, after duplicating many of the best features from several of

Adobe's products, perhaps most notably *Illustrator*. And consider the way *Figma* is now eating into both *Sketch* and *Illustrator*'s market share. Both *Sketch* and *Figma* have done this without needing to make an enormous investment in research. That's a big advantage they have over Adobe.

As Marc Andreessen has observed:

true defensibility purely at the product level is really rare in [Silicon] Valley, because there are a lot of really good engineers... And then there's the issue of leap-frogging. The next team has the opportunity to learn from what you did and then build something better.

Put another way, many tools for thought are public goods. They often cost a lot to develop initially, but it's easy for others to duplicate and improve on them, free riding on the initial investment. While such duplication and improvement is good for our society as a whole, it's bad for the companies that make that initial investment. And so such tools for thought suffer the fate of many public goods: our society collectively underinvests in them, relative to the benefits they provide.

Earlier, we argued that modern design practice generally isn't up to the challenge of producing genuinely transformative tools for thought. On the surface, that process-level argument appears very different to the public goods argument we just made. In fact, the process-level explanation is a consequence of the public goods explanation: companies don't use the necessary processes because there's little value to them in doing so. By contrast, in “harder-tech” industries – say, chip design – companies have much more incentive to do deep research work. In those industries it's considerably harder for other companies to duplicate or capture the value of that research.

It's illuminating to contrast with video games. Game companies develop many genuinely new interface ideas. This perhaps seems surprising, since you'd expect such interface ideas to also suffer from the public goods problem: game designers need to invest enormous effort to develop those interface ideas, and they are often immediately copied (and improved on) by other companies, at little cost. In that sense, they are public goods, and enrich the entire video game ecosystem.

But there's a big difference between video game companies and companies such as Adobe. Many video games make most of their money from the first few months of sales. While other companies can (and do) come in and copy or riff on any new ideas, it often does little to affect revenue from the original game, which has already made most of its money. While this copying is no doubt irritating for the companies being copied, it's still worth it for them to make the up-front investment.

The net result is that in gaming, clever new interface ideas can be distinguishing features which become a game's primary advantage in the marketplace. Indeed, new interface ideas may even help games become classics – consider the many original (at the time) ideas in games ranging from *Space Invaders* to *Wolfenstein 3D* to *Braid* to *Monument Valley*. As a result, rather than underinvesting, many companies make sizeable investments in developing new interface ideas, even though they then become public goods. In this way the video game industry has largely solved the public goods problems.

By contrast, a company like Adobe builds their business around distribution and long-term lock in. They convince people – indeed, entire organizations – to make long-term commitments to their products. Schools offer classes so people can call themselves “*Photoshop* experts” or “*Illustrator* experts”. Companies designate their design departments as “Adobe shops”. So while Adobe does invest in developing clever new interface ideas (for them, unlike the video game companies, this genuinely means tools for thought), it's less central to their competitive advantage, and they invest less than they would if it was their central advantage. And Adobe does perhaps as much or more work developing tools for thought as any company.

It's encouraging that the video game industry can make inroads on the public goods problem. Is there a solution for tools for thought? Unfortunately, the novelty-based short-term revenue approach of the game industry doesn't work. You want people to really master the best new tools for thought, developing virtuoso skill, not spend a few dozen hours (as with most games) getting pretty good, and then moving onto something new.

Another plausible solution to the public goods problem is patents, granting a temporary monopoly over use of an invention. Many software companies, including Adobe, develop a large patent portfolio. However, the current patent system is not a solu-

tion for this problem. In 2017, Dana Rao, Adobe's Vice President for Intellectual Property and Litigation, posted a call for major reforms to the patent system, stating that:

[the patent] system is broken... What happened? A patent gold rush built by patent profiteers... Their value lies not in the innovation behind the patent but in the vagueness of the patent's claims and the ability to enforce it in a plaintiff-friendly forum... Where did the material for these bad patents come from? The advent of software... This led to idea-only patents being granted with broad and often invalid claims, and eager patent profiteers were only too glad to take advantage.

Adobe shares in common with many other software companies that much of their patenting is defensive: they patent ideas so patent trolls cannot sue them for similar ideas. The situation is almost exactly the reverse of what you'd like. Innovative companies can easily be attacked by patent trolls who have made broad and often rather vague claims in a huge portfolio of patents, none of which they've worked out in much detail. But when the innovative companies develop (at much greater cost) and ship a genuinely good new idea, others can often copy the essential core of that idea, while varying it enough to plausibly evade any patent. The patent system is not protecting the right things.

Switching away from the viewpoint of individual companies, and to the viewpoint of society as a whole, not only do we want to incentivize invention, we also want ideas to move reasonably rapidly into the public domain. Think about fundamental tools for thought such as writing and the number system. Obviously, it's good that those spread throughout society, unencumbered by IP concerns! More broadly, many tools of thought become more valuable for society as they become more ubiquitous. Again, here the modern patent system has numerous well-known problems, striking a poor balance between private and public interest. While a well-designed patent system might very well help solve the public goods problem, the patent system we actually have seems poorly adapted to the problem.

Is it possible to avoid the public goods problem altogether? Here's three classes of tools for thought which do:

- Search engines such as Google are tools for thought. They avoid the public goods problem because their value is in their brand and in hard-to-duplicate and capital intensive backend elements (including their data centers, proprietary algorithms, ad network, and distribution), not in their interface ideas.
- A service such as Twitter can be considered a tool for collective thought. While the interface is easily copied, the company is hard to duplicate, due to network effects.
- Novel hardware devices (e.g., for VR, or the Wii remote, or for new musical instruments) can be used as the basis for new tools for thought. While hardware can be duplicated, it's often much more expensive than duplicating software. And, in any case, the advantage for such companies is often in distribution, marketing, and relationships with vendors who make products for the platform.

While these suggestions all avoid the public goods problem, they don't directly solve the public goods problem. And many promising directions – including ideas such as the mnemonic medium and mnemonic videos – involve a substantial public goods element. Is it possible to solve the public goods problem in such cases? The two most promising approaches seem to us to be:

- Philanthropic funding for research. This approach was used, for instance, by the field of computer animation and animated movies. Decades of public research work on computer animation resulted in a large number of powerful and (in many cases) publicly available ideas. This, in turn, helped prepare the way for companies such as Pixar and Dreamworks, which developed many of the ideas further, and took them to scale.
- The model used by Adobe and similar companies, in which new tools for thought are a central part of the company's operations, but not the core of their competitive moat. That moat may instead be built around training, marketing, documentation, and so on.

Questioning our basic premises

There are three important premises we've taken for granted up to now. First is the assertion that we're still in the early days, that many more transformative tools for thought are yet to be discovered. Second is the assertion that work on tools for thought is stalled, that there's not lots of interesting work going on. And third, a kind of meta-premise, is the assertion that this kind of work is worth doing, relative to the current fashion for related ideas such as artificial general intelligence and brain-computer interfaces. In this section we discuss these premises.

What if the best tools for thought have already been discovered?

In other words, perhaps the 1960s and 1970s were an unrepeatable golden age, and all we can expect in the future is gradual incremental improvement, and perhaps the occasional major breakthrough, at a decreasing frequency?

There's a plausible story suggesting this is true. Tech is an enormous industry, well funded, with many bright, ambitious, talented people. Surely if there were major ideas to discover, people would do so? This argument is reinforced by the fact that, at the individual level, we meet many brilliant people who are fascinated by (and often working on) tools for thought, but who nonetheless seem to be making slow progress.

But while this story has a superficial appeal, it's misleading. Really difficult problems – problems like inventing Hindu-Arabic numerals – aren't solved by good intentions and interest alone. A major thing missing is foundational ideas powerful enough to make progress. In the earliest days of a discipline – the proto-disciplinary stage – a few extraordinary people – people like Ivan Sutherland, Doug Engelbart, Alan Kay, and Bret Victor – may be able to make progress. But it's a very bespoke, individual progress, difficult to help others become proficient in, or to scale out to a community. It's not yet really a discipline. What's needed is the development of a powerful praxis, a set of core ideas which are explicit and powerful enough that new people can rapidly assimilate them, and begin to develop their own practice. We're not yet at that stage with tools for thought. But we believe that we're not so far away either.

While that argument is helpful context, it doesn't address the core point: it doesn't mean there are a lot of new transformative tools for thought waiting to be discovered. Again: maybe the most important tools for thought have already been discovered?

We can't predict the future, so it's not possible to answer this question with certainty. But it seems to us that the human race just hasn't really tried very hard yet. When small groups of motivated people do – as in pioneering labs such as PARC, SRI, and other DARPA-inspired early efforts, as well as modern labs such as Dynamicland – they make rapid progress. It's extremely encouraging that those efforts – tiny efforts, in the scheme of humanity's overall research effort – make such rapid progress. To us, that suggests scaling them up, becoming much more ambitious.

Why not work on artificial general intelligence (AGI) or brain-computer interfaces (BCI) instead?

We're often asked: why don't you work on AGI or BCI instead of tools for thought? Aren't those more important and more exciting? And for AGI, in particular, many of the skills required seem related.

They certainly are important and exciting subjects. What's more, at present AGI and BCI are far more fashionable (and better funded). As a reader, you may be rolling your eyes, supposing our thinking here is pre-determined: we wouldn't be writing this essay if we didn't favor work on tools for thought. But these are questions we've wrestled hard with in deciding how to spend our own lives. One of us wrote a book about artificial intelligence before deciding to focus primarily on tools for thought; it was not a decision made lightly, and it's one he revisits from time to time. Indeed, given the ongoing excitement about AGI and BCI, it would be surprising if people working on tools for thought didn't regularly have a little voice inside their head saying “hey, shouldn't you be over there instead?” Fashion is seductive.

One striking difference is that AGI and BCI are based on relatively specific, well-defined goals. By contrast, work on tools for thought is much less clearly defined. For the most part we can't point to well-defined, long-range goals; rather, we have long-

range visions and aspirations, almost evocations. The work is really about exploration of an open-ended question: how can we develop tools that change and expand the range of thoughts human beings can think?

Culturally, tech is dominated by an engineering, goal-driven mindset. It's much easier to set KPIs, evaluate OKRs, and manage deliverables, when you have a very specific end-goal in mind. And so it's perhaps not surprising that tech culture is much more sympathetic to AGI and BCI as overall programs of work.

But historically it's not the case that humanity's biggest breakthroughs have come about in this goal-driven way. The creation of language – the ur tool for thought – is perhaps the most important occurrence of humanity's existence. And although the origin of language is hotly debated and uncertain, it seems extremely unlikely to have been the result of a goal-driven process. It's amusing to try imagining some prehistoric quarterly OKRs leading to the development of language. What sort of goals could one possibly set? Perhaps a quota of new irregular verbs? It's inconceivable!

Similarly, the invention of other tools for thought – writing, the printing press, and so on – are among our greatest ever breakthroughs. And, as far as we know, all emerged primarily out of open-ended exploration, not in a primarily goal-driven way. Even the computer itself came out of an exploration that would be regarded as ridiculously speculative and poorly-defined in tech today. Someone didn't sit down and think “I need to invent the computer”; that's not a thought they had any frame of reference for. Rather, pioneers such as Alan Turing and Alonzo Church were exploring extremely basic and fundamental (and seemingly esoteric) questions about logic, mathematics, and the nature of what is provable. Out of those explorations the idea of a computer emerged, after many years; it was a discovered concept, not a goal. Fundamental, open-ended questions seem to be at least as good a source of breakthroughs as goals, no matter how ambitious. This is difficult to imagine or convince others of in Silicon Valley's goal-driven culture. Indeed, we ourselves feel the attraction of a goal-driven culture. But empirically open-ended exploration can be just as, or more successful.

“What will new tools for thought be like?” is a question we hear often. And yet, almost by definition, we cannot say. As we noted earlier, if we could communicate the experience in an essay, then the tools would be failing at their job; they would not be transforming a person's thinking, or even their consciousness. Concretely: to understand the mnemonic medium you must use it intensively over an extended period. And even then you may not be conscious of the effect; we've done interviews with users who are apparently unaware of the incredible level of recall they have of material in the essay they have read. One of the most famous papers in the philosophy of consciousness is entitled “What is it like to be a bat?” Each tool for thought poses a similar question, near impossible to answer without immersion in the tool: “What is it like to be a language user? A musician?” and so on.

It seems plausible to us that work on tools for thought will be, over the next few decades, more important than work on AGI and BCI. And, given how fashionable and well-funded work on AGI and BCI currently is, it seems nearly certain that work on tools for thought offers vastly greater benefit, at the margin.

What about the longer term? There, the situation is less clear. It seems likely that the three fields will merge, or at least feed strongly into one another. Together with Shan Carter, one of us has argued that one of the most promising applications for AI is as a way of discovering new tools for thought.

BCI seems likely to be even more closely related. BCI is sometimes described using ideas like a memory chip for long-term memories, or some way of increasing short-term working memory. Such ideas may well become important. But it also seems possible that BCIs will be used to enable new mental operations, new mental representations, and new affordances for thought; in short, the same kind of things as are involved in developing non-BCI tools for thought. Perhaps we'll develop the capacity to directly imagine ourselves in 4 or 5 or more dimensions; or traversing a Riemann manifold; or the ability to have multiple tracks of conscious attention. These are about changing the interface for thought, the basic abstractions and operations which are allowed. And so it seems plausible that work today on tools for thought will directly impact the way we use BCIs in the future.

Tools for thought must be developed while doing serious work. The aspiration to canonical content

Seymour Papert, one of the principal creators of the Logo programming language, had a remarkable aspiration for Logo. Logo is sometimes described as a “programming language for children”, and people sometimes think Papert was mostly interested in helping children learn how to program. But that wasn't Papert's principal intent. Rather, Papert wanted to create an immersive environment – a kind of “Mathland” – in which children could be immersed in mathematical ideas. In essence, children could learn differential geometry by going to Mathland.

It's a beautiful aspiration, and Logo contains many striking ideas. But as far as we know, no professional differential geometer (or, more generally, mathematician) uses Logo seriously as a tool in their work. And upon reflection that seems troubling. If Logo genuinely expresses the ideas of differential geometry, why don't differential geometers use it? You start to wonder: might it be that Logo leaves out important ideas about differential geometry, maybe even the most important ideas about differential geometry? After all, while mathematically trained, Papert wasn't himself an accomplished differential geometer. How would he even have known what to include? And certainly most of the people interested in Logo aren't qualified to make that judgment.

There's a standard retort to this, which we've heard from within the Logo community. It's to talk about the “floor” and “ceiling” of different environments for thought. In this account, Logo has a low floor (meaning anyone can use it) and a low ceiling (so it's not well adapted for the sort of advanced work a professional would want to do).

At first this sounds plausible. But upon reflection it's difficult to make much sense of. How do the creators of Logo know that mastering Logo helps later with understanding real (forgive us!) differential geometry? What's the criterion for success? One of us (MN) worked for several years doing research in the closely related field of Riemannian geometry. While Logo is enjoyable to use and contains many fun ideas, MN has trouble seeing that learning Logo would help much in learning differential geometry.

At the end of Norvig's economics essay is a short afterword explaining how he came to write the essay. Shortly before writing the essay he'd heard about the kinds of economic models discussed in the notebook, and he wanted to explore several questions about them. After talking it over with some colleagues they decided to each independently attack the problems, and to compare notes. Although Norvig's essay is, in some sense, “educational”, Norvig's intent was to explore a set of problems he himself was genuinely curious about. The educational aspect was a byproduct.

And so what you have is a world-class research scientist who wanted to explore a set of questions. He used the Jupyter medium to do those explorations, and then to share that exploration with the world. And he shared it in a form where others could immediately build upon and extend his thinking.

There's a lot of work on tools for thought that takes the form of toys, or “educational” environments. Tools for writing that aren't used by actual writers. Tools for mathematics that aren't used by actual mathematicians. And so on. Even though the creators of such tools have good intentions, it's difficult not to be suspicious of this pattern. It's very easy to slip into a cargo cult mode, doing work that seems (say) mathematical, but which actually avoids engagement with the heart of the subject. Often the creators of these toys have not ever done serious original work in the subjects for which they are supposedly building tools. How can they know what needs to be included?

Concretely: suppose you want to build tools for subject X (say $X =$ differential geometry). Unless you are deeply involved in practicing that subject, it's going to be extremely difficult to build good tools. It'll be much like trying to build new tools for carpentry without actually doing any carpentry yourself. This is perhaps part of why tools like *Mathematica* work quite well – the principal designer, Stephen Wolfram, has genuine research interests in mathematics and physics. Of course, not all parts of *Mathematica* work equally well; some parts feel like toys, and it seems likely those are the ones *not* being used seriously internal to the company.

There's a general principle here: *good tools for thought arise mostly as a byproduct of doing original work on serious problems*. They tend either be created by the people doing that work, or by people working very closely to them, people who are genuinely bought

in. Furthermore, the problems themselves are typically of intense personal interest to the problem-solvers. They're not working on the problem for a paycheck; they're working on it because they desperately want to know the answer.

Many people have asked why we wrote our first mnemonic essay about quantum computing. If we'd chosen an easier subject we could have attracted a much larger audience. But we also wanted the essay to be authentic, to be about problems we wanted to solve. One of us (MN) has done a lot of original research work on quantum computing. The essay reflects that thinking. Indeed, the framing of the essay is about answering a question MN personally wanted to answer: if humans ever discovered aliens, would they have computers, and if so, what types of computer would they have? This perhaps sounds like a contrived question, but it's quite serious, and turns out to be a deep question with a nontrivial answer: writing the essay helped MN substantially improve his understanding of the question.

That said, answering that question wasn't the principal point of creating the essay: making the mnemonic medium was. And for future work on tools for thought, it'd be valuable to push much harder on questions we'd genuinely like to answer ourselves. That's a way of keeping yourself honest, ensuring you're not just building a flashy toy, but something genuinely useful for solving real problems that are of independent interest.

In serious mediums, there's a notion of *canonical media*. By this, we mean instances of the medium that expand its range, and set a new standard widely known amongst creators in that medium. For instance, *Citizen Kane*, *The Godfather*, and *2001* all expanded the range of film, and inspired later film makers. It's also true in new media. YouTubers like Grant Sanderson have created canonical videos: they expand the range of what people think is possible in the video form. And something like the *Feynman Lectures on Physics* does it for textbooks. In each case one gets the sense of people deeply committed to what they're doing. In many of his lectures it's obvious that Feynman isn't just educating: he's reporting the results of a lifelong personal obsession with understanding how the world works. It's thrilling, and it expands the form.

We've been disappointed by how unambitious people are in this sense with *Jupyter* notebooks. They haven't pushed the medium all that hard; there is no *Citizen Kane* of Jupyter notebooks. Indeed, we're barely beyond the Lumière brothers. Examples like Norvig's notebook are fine work, but seem disappointing when evaluated as leading examples of the medium.

Aspiring to canonicity, one fun project would be to take the most recent IPCC climate assessment report (perhaps starting with a small part), and develop a version which is executable. Instead of a report full of assertions and references, you'd have a live climate model – actually, many interrelated models – for people to explore. If it was good enough, people would teach classes from it; if it was really superb, not only would they teach classes from it, it could perhaps become the creative working environment for many climate scientists.

One promising exploration in this direction is *The Structure and Interpretation of Classical Mechanics*, a beautiful executable book building up classical mechanics. Many theorems of classical mechanics aren't just expressed in static form, on the page, but live, as code which can be modified by the user. Theorems become APIs, which can literally be applied to other objects, and chained together. It uses a much more powerful underlying model than Jupyter, developing a new symbolic language as part of the book. It has many flaws – among them, the book doesn't run live in the browser, making it difficult for users to experiment. And while the book is well written, the authors do not understand classical mechanics as deeply as the authors of some other books. But it's nonetheless an inspiring evocation of what is possible. And it hints at what is possible when authors use executable books for a serious purpose, and aspire toward canonical media.

Let's return to the question that began the essay: how to build transformative tools for thought? Of course, we haven't even precisely defined what such transformative tools are! But they're the kind of tools where relatively low cost changes in practice produce transformative changes in outcome – non-linear returns and qualitative shifts in thinking. This is in contrast with the usual situation, where a small change in practice causes a small change in results.

Historically, humans have invented many such transformative tools for thought. Writing and music are ancient examples; in modern times, tools such as *Photoshop* and *AutoCAD* qualify. Although it's very early days, we believe the mnemonic medium shows much promise. It needs to be developed much further, along the lines we've described, and likely requires additional powerful ideas. But we believe it's possible for humanity to have a widespread memory practice that radically changes the way we think.